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Title of the Experiment:

Heart Rate of 20 Burpees vs One Sprint

Background Information:

- **Describe** the validity of the research question be tested by scientific investigation
- **Explain** the reasoning behind the hypothesis using correct scientific reasoning

Validity of the Research Question:

The purpose of this research is to figure out which would have a faster recovery, two sprints across the distance of 23.5 meters or twenty burpees, and to do that, the heart rate would be recorded to see which would have a higher heart rate afterwards. And whatever the result is, the opposite would be the ideal option for faster recovery.

Reasoning behind the Hypothesis:

The reason there would be a lower heart rate after two sprints, rather than twenty burpees, is that burpees contain much more unique movements (Lawler), whereas running is a common activity for the majority.

Research Question:

- Must be based on the **description** given in the Background Information

What would cause a higher heart rate: 20 burpees, or two sprints?

Hypothesis:

- Must be based on the reasoning given in the Background Information
- **State** the hypothesis in an "if, then, because" format
- Must include all variables used, followed by a chosen variable

If two sprints were performed, then it would have a lower heart rate, because burpees involve more varied movements.

Variables:

- **Identify** the dependent variable and the independent variable
- **Describe** how the data for each variable will be collected

Variable	The variable is:	How the data will be collected
Dependent	Heart rate	Smart watch
Independent	Amount of burpees	Smart watch

Control Variables:

- **Identify** each control variable (**add more rows to the table if needed; not all rows need to be used**)
- **Describe** the method used to allow each variable to be controlled

Variable	Method of Control
The heart rate measurer	Using my classmates' smart watch every time
The distance of the basketball court.	Measuring the length (23.5 meters)
Amount of burpees.	Counting the number of burpees
Four test subjects	Four people, two boys and two girls

Materials:

- Should be in a list (not a paragraph); you can bullet the list
- Should be as specific as possible (ie. don't just say "beakers," say Ten 250 mL beakers, for example)

- **4 test subjects (2 male and 2 female)**
- **The basketball court outside the grammar school (23.5 meters)**
- **The Series 7 Apple Watch has a heart rate app**
- **A pencil**

- **A piece of paper**

Method:

- Should be written in the past tense
- Should be numbered - do not use bullet points
- Should be clear and concise - **repeat steps** should be used wherever possible
- Should be logical, safe, and complete

1. Test subjects 1, 2, 3, and 4 were found
2. A pencil and a piece of paper were obtained
3. Test subjects 3 and 4 were put aside for now
4. Test subject 1 had put on the Series 7 Apple Watch with the heart rate app open
5. Test subject 1 was then prepared to run across the basketball court and back
6. Once the test subject 2 said go, test subject 1 ran across the basketball court and back
7. Test subject 4 then wrote down on the piece of paper with a pencil what the heart rate was and which test subject after running across the basketball court and back
8. Test subject 1 gave test subject 4 the Series 7 Apple Watch
9. Steps 5-7 were then repeated for test subject 4
10. Test subject 4 gave test subject 2 the Series 7 Apple Watch
11. Test subject 2 was prepared to do 20 non-push-up burpees
12. The test subject did 20 non-push-up burpees
13. Test subject 1 wrote down the heart rate on the piece of paper

14. Steps 10-13 were repeated for test subject 3

15. Test subjects 1 and 2 heart rates were compared

16. Step 15 was repeated for test subjects 3 and 4